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Reciprocal-space instrumental effects on the real-space neutron atomic pair distribution function

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An atomic pair distribution function (PDF) neutron powder diffraction round-robin experiment was performed on six diffractometers at three spallation sources. Instrument-specific effects on the real-space PDF were investigated, such as finite measurement range, the instrument resolution and the asymmetric shape of diffraction peaks. Two illustrative samples, a perfectly long-range-ordered element, Pb, and a locally strained alloy $ZnSe_{0.5}Te_{0.5}$, were measured at low temperatures. Various aspects of the PDF were explored, either qualitatively by direct comparison or quantitatively *via* structural modelling. Future implementation of modelling codes incorporating some of these instrumental effects are also discussed.

1. Introduction

The real-space atomic pair distribution function (PDF) technique specializes in high-resolution local-structure determination (Egami & Billinge, 2003; Proffen *et al.*, 2003). Traditionally it has been applied to study glasses and amorphous materials; however, more recently it has successfully been applied to study disorder in crystals. It can also obtain the average structure as a complement to conventional crystallographic analysis (Egami & Billinge, 2003). PDF measurement of the total structure is made possible by collecting both the Bragg intensity and diffuse scattering over a wide range of momentum transfer. The PDF technique has been shown to provide unique insights into the local and average structure of materials with various degrees of structural complexity (Billinge & Kanatzidis, 2003; Tucker *et al.*, 2000; Hibble & Hannon, 2002; Billinge *et al.*, 1996; Louca *et al.*, 1997; Božin *et al.*, 2000; Petkov *et al.*, 2002; Mamontov & Egami, 2000). This real-space local-structure probe also opens up the possibility of applying scattering techniques to study increasingly disordered materials of growing interest, which challenge the conventional average-structure approach (Egami & Billinge, 2003).

Various neutron powder diffractometers have been intensively used for recent PDF studies. The resulting experimental PDFs are inevitably subject to their different instrument characteristics. These effects are small and often are neglected during the data analysis. Nonetheless, to achieve greater accuracy it is necessary to characterize and account for these effects. We present here the result from the first PDF neutron powder diffractometer round-robin experiment. Instrument specific effects, such as the finite measurement range, and various aspects of the complicated instrument resolution function, are investigated.

2. Method

Two samples were chosen for this round-robin experiment. One is the ternary alloy compound $ZnSe_{0.5}Te_{0.5}$ with zinc-blende structure and local atomic strains. The method of sample preparation can be found elsewhere (Peterson *et al.*, 2001). The other is the simple element lead with a perfect face-centred cubic (f.c.c.) lattice. Pb powders (200 mesh, metal basis 99%) were purchased from Alfa Aesar (stock No. 00942) and used as received. Six neutron powder diffractometers at three spallation sources were selected. They are the NPDF and NPD at the Lujan Centre at Los Alamos National Laboratory, the GPPD, SEPD and GLAD at the Intense Pulsed Neutron Source at Argonne National Laboratory, and the GEM at ISIS at the Rutherford Appleton Laboratory, UK.[1]

Finely powdered samples of $ZnSe_{0.5}Te_{0.5}$ (8.44 g) and Pb (28.37 g) were packed into cylindrical vanadium cans with helium exchange gas. The sample can used has a nominal diameter of 3/8 inch, inner radius of 0.4636 cm, outer radius 0.4763 cm, and height of around 5.4 cm. The same $ZnSe_{0.5}Te_{0.5}$ sample was used during all experiments, and Pb powders were taken from the same bottle.

All experiments followed a typical PDF set methodology (Egami & Billinge, 2003). For example, beam collimators were chosen to achieve the largest effective scattering volume and detector banks may be regrouped to achieve better statistics. Data collection was carried out at low temperatures to mini-

[1] NPDF: the Neutron Powder Diffractometer/Neutron Pair Distribution Function Diffractometer (newly upgraded in 2002); NPD: Neutron Powder Diffractometer, the same as NPDF before it was upgraded; GPPD: the General Purpose Powder Diffractometer (before its recent upgrade); SEPD: the Special Environment Powder Diffractometer; GLAD: the Glass, Liquid and Amorphous Materials Diffractometer; GEM: the General Materials Diffractometer.

Table 1
Temperature and data collection times for $ZnSe_{0.5}Te_{0.5}$ and Pb samples on the six diffractometers used in this study.

	$ZnSe_{0.5}Te_{0.5}$		Pb	
Instrument	T (K)	Time (h)	T (K)	Time (h)
NPDF	15.0	1.25	15.0	5
GPPD	10.0	12	10.0	24
NPD	30.0	12	12.0	10
SEPD	10.0	10	10.0	12
GLAD	10.0	20	10.0	24
GEM	10.0†	0.5†	20.0	2.5

† These data were collected before the back-scattering detector module was installed on GEM.

mize the effects of thermal motion. In each case the sample can was mounted on the cold finger of a closed-cycle Helium refrigerator. Table 1 lists detailed sample temperatures and collection times for each experiment. It is worth pointing out that data collection time during some experiments is excessively long for other purposes. For strongly scattering samples, experience suggests that 4–6 h should be sufficient for SEPD, GLAD may require 8–10 h, while data collection on GEM or NPDF requires 0.5–1 h per data set.

The empty displex, an empty container mounted on the displex and the empty instrument, were all measured, allowing us to assess and subtract instrumental backgrounds. The scattering from a vanadium rod was also measured to normalize with respect to the incident spectrum and detector efficiencies. Standard data corrections were carried out as described elsewhere (Egami & Billinge, 2003) using the program *PDFgetN* (Peterson *et al.*, 2000). After being corrected, the data were normalized by the total scattering cross section of the sample to yield the total scattering structure function, $S(Q)$, where Q is the magnitude of the scattering vector. This was then converted to the PDF, $G(r)$, by a Sine Fourier transform according to $G(r) = (2/\pi)\int_0^\infty Q[S(Q)-1]\sin Qr\,\mathrm{d}Q$. Q_{max} of 32.0 Å was used for GLAD, while a value of 38.0 $Å^{-1}$ was used for the other instruments, which is a typical value used for PDF studies on those diffractometers. Both NPDF and GEM are capable of higher Q_{max}; however, 38.0 $Å^{-1}$ was chosen to allow comparison of all the diffractometers at the same Q_{max}. GLAD spectra had to be limited to a Q_{max} of 32 $Å^{-1}$ due to the appearance of unphysical features at higher values of Q.

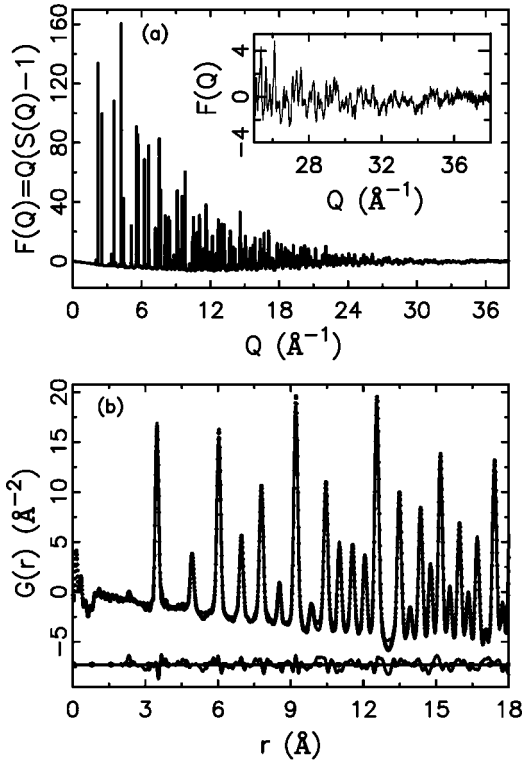


Figure 1
(*a*) Reduced structure function $F(Q)$ of Pb from NPDF with Q_{max} of 38.0 $Å^{-1}$. The inset shows the high-Q region on an expanded scale, noting the presence of rich structural information. (*b*) The experimental $G(r)$ (dots) obtained by Fourier transformation of the $F(Q)$ and the calculated PDF from the refined structural model (solid line). The difference curve is shown offset below.

Structure refinements with Pb PDFs were carried out using *PDFFIT* (Proffen & Billinge, 1999), a real-space full-profile non-linear-regression fitting program. Conventional Rietveld refinement of Pb was performed using the program *GSAS* (Larson & Von Dreele, 2000). Fig. 1(*a*) shows the reduced structure function, $F(Q) = Q[S(Q) - 1]$, of Pb data from NPDF. The $G(r)$ and the PDF calculated from the refined model are shown in Fig. 1(*b*) with a difference curve shown offset below. The PDF of this f.c.c. metal is clearly of high quality.

3. Results and discussion

3.1. Effect of finite measurement range

The effect of a finite measurement range on the PDF is easily understood using the convolution theorem. The PDF of the experimental data can be thought of as the ideal PDF of the structure being measured, convoluted with a termination function that is the Fourier transform of a step function that terminates the data at Q_{max} in reciprocal space. The step function has the form $\Theta(Q) = \{1, \text{for } Q \leq Q_{max}; 0, \text{for } Q > Q_{max}\}$, which has Fourier transform $\theta(r) = (Q_{max}/\pi)j_0(Q_{max}r)$, where $j_0(x)$ denotes the zeroth order spherical Bessel function (Thorpe *et al.*, 1998). The old practice of damping the Q-space data in the high-Q region with a smooth function to reduce termination effects in the Fourier transform is not necessary when data are measured to a sufficiently high Q_{max}, as is evident in Fig. 1(*b*). The convolution of the PDF with function $\theta(r)$ results in Gaussian-like peak broadenings and the so-called termination ripples oscillating away from peak positions. Qualitatively, the magnitude of the peak broadening and the amplitude of the termination ripples increases monotonously with decreasing Q_{max}. For a single Gaussian PDF peak with intrinsic width σ_U and unit height, its height after convolution is given by $\mathrm{erf}\,(2^{1/2}\pi Q_{max}\sigma_U)$, where the error function $\mathrm{erf}\,(x) = (2/\pi^{1/2})\int_0^x \exp(-u^2)\,\mathrm{d}u$. The magnitude of Q_{max} is the most important factor determining the PDF real-space resolution.

A good example to demonstrate the effect of Q_{max} is the compound $ZnSe_{0.5}Te_{0.5}$, which contains two distinct nearest-

neighbour bonds, Zn—Se and Zn—Te, differing by just 0.14 Å. Fig. 2(*a*) shows that only one single peak is observed when Q_{max} = 25.0 Å^{-1}. This doublet becomes better resolved when Q_{max} is increased, as is evident in Fig. 2(*a*). Ripples around the base of the peak also become evident at higher Q_{max}. These are not termination ripples, but arise because of increasing amounts of random noise in the data. As Q_{max} increases, the signal in the data decreases due to Debye–Waller effects, but the statistical noise in the data increases. Extending to large Q_{max} thus leads to increased noise in the PDF curves.

For the nearest-neighbour single peak of Pb at 3.48 Å, the increment of peak height (shown in Fig. 2*b*) with increasing Q_{max} is clear, as is the diminishing of the termination ripples. The peak height as a function of Q_{max} is shown in Fig. 2(*c*), where good consistency can be seen on different instruments. The plot of peak height *versus* Q_{max} saturates around Q_{max} = 37.0 Å^{-1}. At this point the resolution is becoming sample-limited rather than measurement-limited. No extra information is being added to the Fourier transform by extending Q_{max} above this point, at least regarding this particular PDF peak. This peak, which corresponds to the nearest neighbours in Pb, is the sharpest PDF peak (Jeong *et al.*, 1999) and so it is most sensitive to the high-Q part of the spectrum. Since it does not change it is safe to say that in this case no structural information is contained in the scattering data above ~37.0 Å^{-1}.

3.2. Effect of instrument resolution function

Here we investigate how the PDF is affected by the diffractometer instrument resolution function (IRF). For time-of-flight (TOF) neutron powder diffractometers, this function has a complicated form that is a convolution of rising and falling exponentials with Gaussian and Lorentzian contributions (Larson & Von Dreele, 2000). The peaks are asymmetric and their widths depend on Q. To understand what aberrations these effects will cause to the experimentally determined PDF, it is helpful to consider simplified forms of the resolution function. In the next section we discuss the predictions of these various approximations to the IRF. In §3.2.2 we investigate how they correspond to the actual experimental observations.

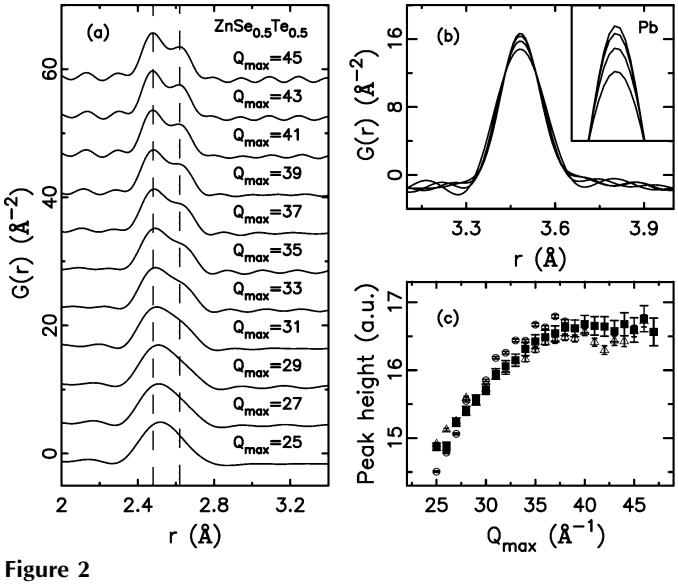


Figure 2
(*a*) The nearest-neighbour peaks of the $ZnSe_{0.5}Te_{0.5}$ PDFs from NPDF with Q_{max} from 25.0 to 45.0 Å^{-1}. The two vertical lines indicate the two distinct bonds at 2.48 (Zn—Se) and 2.62 Å (Zn—Te). (*b*) The nearest-neighbour peak at 3.48 Å of the Pb PDFs from NPDF with Q_{max} of 25.0, 30.0, 35.0 and 45.0 Å^{-1}. The inset shows the top part of the peaks on an expanded scale. (*c*) The height of the Pb PDF peak shown in (*b*) plotted *versus* Q_{max}. Data from NPDF (squares), GEM (triangles) and SEPD (circles) are shown and scaled by their respective scale factors. Error bars correspond to the standard deviation.

3.2.1. Functional analysis. The simplest approximation is that the IRF is a single Gaussian of constant width σ_q. The Fourier transform of this is a Gaussian in real space with a width of $1/\sigma_q$. Since the Q-space data are convoluted by the IRF, the experimental PDF is the product of this real-space Gaussian function and the ideal PDF obtained with infinite instrument resolution. Thus we expect that the amplitudes of the peaks will be diminished, following the Gaussian form of

$$S(r) = \exp(-\sigma_q^2 r^2/2),$$

where S is the r-dependent scale factor. In fact, the resolution of TOF diffraction measurements depends linearly on Q. The implied lower resolution at higher Q is not much welcome because of the increasing overlapping of the diffraction peaks with Q. This Q dependence complicates the simple application of the convolution theorem. In general there is not an analytic solution for the effect of a Q-dependent resolution on the PDF. However, in the special case of $\sigma_q^2(Q) = \sigma_{q0}^2 + \alpha^2 Q^2$, an analytic solution does exist (Thorpe *et al.*, 2002). The result is that, besides the damping effect described above, there is an additional, r-dependent broadening of peaks in the PDF given by

$$\sigma_r^2(r) = \sigma_U^2 + \alpha^2 r^2, \qquad (1)$$

where $\sigma_U = (U_{PDF})^{1/2}$ corresponds to the amplitude of the thermal and quantum vibrations in the specimen (Thorpe *et al.*, 2002). The considered form of $\sigma_q(Q)$ converges to the measured ($\Delta Q \propto Q$) behaviour in the IRF and it gives a physical explanation for the earlier observations that an additional, r^2-dependent broadening of the PDF peaks significantly improves data fits (Toby & Egami, 1992). The origin of this r-space broadening can be qualitatively explained in the following way. At high-r PDF, the diffraction data are being sampled by shorter wavelength sine waves. When the sampling wave has a wavelength comparable with the peak width, the peak intensity is cancelled and does not contribute to the Fourier transform. If diffraction peaks are becoming broader with increasing Q, then their intensity is being progressively lost with increasing Q, similar to an effective additional component to the Debye–Waller factor, resulting in an artificial broadening of the PDF peaks. This artificial broadening is more pronounced as r increases and the sine-wave frequency becomes smaller (Billinge, 1992).

The next factor to consider is the fact that the IRF has asymmetric peak shapes. In TOF measurements, the leading rising exponential of the peak is sharper than its trailing declining counterpart (the same holds for the d scale; however

on a Q scale the leading exponential is broader). To determine lattice parameters accurately it is necessary to convert TOF to d spacing or Q, which is typically done using diffractometer constants that are refined against calibration data of known lattice parameter using Rietveld refinement. The fitting procedure of the Rietveld refinement assigns the peak position not at the position of maximum intensity but displaced toward the sharp high-Q trailing edge of the peak. Because of self-consistency this does not present a problem when the same profile function will be used to refine a data set of unknown lattice parameter using the Rietveld method. However, the way the Fourier transform samples the peaks is different. If, as is commonly done, Rietveld-derived diffractometer constants are used for the TOF-to-Q conversion in PDF studies, a systematic error results in PDF-refined lattice parameters. Because of the Q dependence of the peak widths, this could also result in an r dependence of the refined lattice parameter. Exactly how this happens is not known in detail, but it is investigated empirically here.

3.2.2. Data analysis. As discussed qualitatively, a Gaussian IRF will cause the PDF to lose intensity with increasing r. Lower resolution data will be more heavily damped. To assess the resolution of the data, a doublet in the diffraction pattern of Pb is shown in Fig. 3(*a*) from all six diffractometers in the order of high (NPDF/GEM/GPPD) to low (GLAD) Q-space resolution from top to bottom. Fig. 3(*b*) shows the low-r region of the corresponding PDFs, where only small differences in PDF peak intensities can be identified. The high-r region of the PDFs in Fig. 3(*c*) shows a clear correlation between the instrument resolution and the PDF intensity. Lower instrumental resolution causes the PDF intensity to diminish faster.

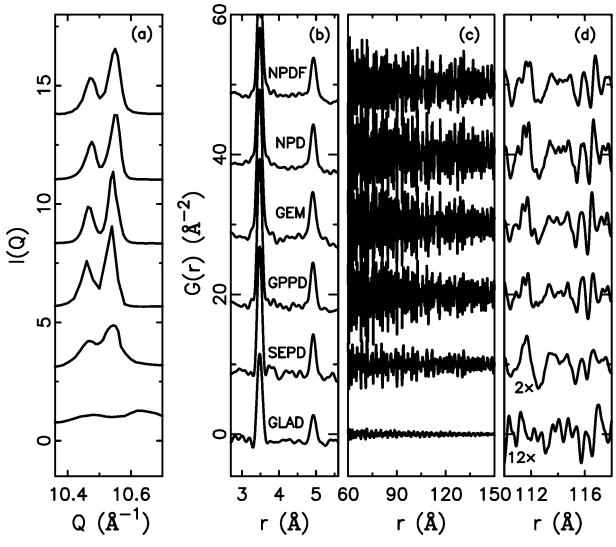


Figure 3
Experimental data shown in all frames are ordered as NPDF, NPD, GEM, GPPD, SEPD and GLAD (from top to bottom). GLAD data have Q_{max} of 32.0 Å^{-1}, while the others have Q_{max} of 38.0 Å^{-1}. (*a*) A doublet near Q = 10.50 Å^{-1} taken from the diffraction pattern of Pb. (*b*) Low-r region of all experimental Pb PDFs where the differences are very small. (*c*) High-r region of the experimental Pb PDFs where the differences are rather significant. (*d*) A focused region of the experimental Pb PDFs. Note that SEPD and GLAD data are amplified by 2 and 12, respectively.

We would also like to point out the reproducibility of the high-r PDF peaks, as demonstrated in Fig. 3(*d*). The SEPD and GLAD data are magnified for the purpose of illustration. This shows that the structural information in the PDF on the nanometre length-scale (in this case at ∼10 nm) is quantitatively reliable. The one exception may be the GLAD diffractometer where the low resolution has led to a loss of intensity and poor signal to noise in this region.

The results are qualitatively similar for $ZnSe_{0.5}Te_{0.5}$. The nearest-neighbour peaks of $ZnSe_{0.5}Te_{0.5}$ from all six diffractometers are shown in Fig. 4(*b*), while the reduced structure functions $F(Q)$ of $ZnSe_{0.5}Te_{0.5}$ are shown in Fig. 4(*a*). It is worth noting the presence of diffuse scattering intensities at high Q [inset to Fig. 4(*a*) where the NPDF data after moderate smoothing are plotted on an expanded scale; the fitted sine wave is a guide to the eye]. This illustrates that, in this sample, important structural information is contained in the data at least to 38 Å^{-1}. The nearest-neighbour doublet in the PDF is clearly better resolved in the Q_{max} = 38 Å^{-1} data than with Q_{max} = 32 Å^{-1}.

On the other hand, Q-space instrumental resolution has little effect on real-space resolution at these nearest-neighbour distances. At the same Q_{max} of 38.0 Å^{-1}, the SEPD appears to resolve this doublet (indicated by the two dashed vertical lines) equally well as NPDF and GEM, even with considerably lower instrument resolution. At the Q_{max} of 32.0 Å^{-1}, GLAD appears to do as well as NPDF, which has significantly higher reciprocal-space resolution. For near-neighbour peaks, real-space resolution is not affected by

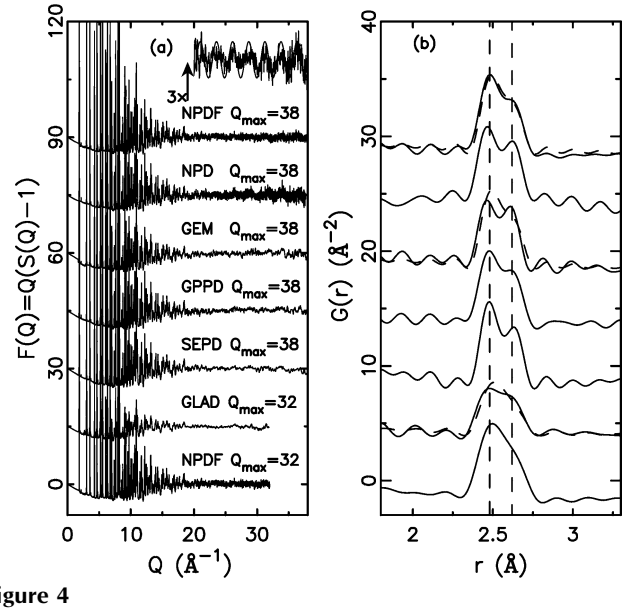


Figure 4
(*a*) The reduced structure function $F(Q)$ of sample $ZnSe_{0.5}Te_{0.5}$ from NPDF, GPPD, GEM, NPD and SEPD with Q_{max} of 38.0 Å^{-1}, as well as from NPDF and GLAD with Q_{max} of 32.0 Å^{-1}. The inset shows the expanded high-Q NPDF data and reveals the signal from diffuse scattering, which is marked by a solid guide line. (*b*) Low-r region of the corresponding $ZnSe_{0.5}Te_{0.5}$ $G(r)$ from each $F(Q)$ shown in (*a*). Calculated PDFs from the distorted lattice structure with Q_{max} of 38.0 and 32.0 Å^{-1} are plotted on top of NPDF and GEM $G(r)$ data for each respective Q_{max}. The two dashed vertical lines are shown at r of 2.48 and 2.62 Å to indicate the positions of Zn—Se and Zn—Te peaks, respectively.

Table 2
Lattice parameters and temperature coefficients obtained from Rietveld refinement and PDF analysis, along with the slope of scale factor decay, m_r.

Instrument	T (K)	a_{Rv} (Å)	a_{PDF} (Å)	U_{Rv} (10^{-2} Å^2)	U_{PDF} (10^{-2} Å^2)	α	m_r (Å^{-1})
GLAD	10	–	4.9143	–	0.263	0.00225	0.0058
SEPD	10	4.9106	4.9149	0.175	0.261	0.00094	0.0036
GPPD	10	4.9146	4.9171	0.331	0.415	0.00078	0.0047
NPD	12	4.9158	4.9201	0.460	0.291	0.00106	0.0007
NPDF	15	4.9230	4.9249	0.482	0.402	0.00108	0.0011
GEM	–	4.9136	4.9148	0.385	0.392	0.00077	0.0044

reciprocal-space resolution, though this becomes less true with increasing r, as we discuss below.

Superimposed on the data in Fig. 4(b) as dashed lines are PDFs calculated from models of the $ZnSe_{0.5}Te_{0.5}$ structure including the alloy disorder (Peterson *et al.*, 2001). The model curves reveal that the double-peak is correctly represented by a higher left peak with a shoulder on the right as seen in the NPDF data. The impression of two peaks separated by a valley in the GEM and SEPD data can be traced to noise ripples that modulate the shape of the peak. These will not affect physical models fit over extended ranges of r, but emphasize that discretion should be exercised when ascribing significance to specific features on individual peaks.

As discussed above, in the simplest case of a Gaussian IRF we expect a Gaussian fall-off in peak amplitude in the PDF. This fall-off is investigated quantitatively for the Pb data from each instrument in Fig. 5. The elemental Pb compound has a perfect f.c.c. lattice with only a few refinable parameters; thus it presents a good measure of the PDF peak profiles. Here PDF refinements have been carried out over 12 Å ranges ('box-cars') where the centroid of the refinement range is systematically shifted to higher r in 5 Å steps. The overall refinement range is from 3.0 to 150.0 Å. Lattice parameter (a), scale factor (S) and isotropic thermal displacement factor (U) are refined. Instrumental parameters in *PDFFIT* that give an r dependence to peak intensities or widths, such as σ_Q, are turned off so any r dependence is immediately evident in a, S and U.

The agreement between the experimental and refined PDFs is very good, evident from the R_{wp} factor shown in Fig. 5(b). While the mean square of the difference actually decreases with r (shown in Fig. 5c), R_{wp} increases as the box moves to higher r, due to the decreased magnitude of $G(r)$.

Fig. 5(a) shows the r dependence of the scale factor and thus the damping. To evaluate the functional form of the fall-off in intensity we have fitted Gaussian, Lorentzian and linear functions. Gaussian and Lorentzian might be expected if the peak shapes in Q space are predominantly Gaussian or exponential, respectively. Surprisingly, neither form does as well as a simple linear fall-off. The linear fits give lower residuals for all of the six diffractometers. This behaviour is reminiscent of $1/f$ noise for fluctuations in solids (Dutta & Horn, 1981). In that case the $1/f$ behaviour in the Fourier space results from a broad uniform distribution of fluctuation frequencies. The analog in the present case may come about because there is a uniform distribution of peak widths in Q space due to the $\sigma_q \propto Q$ resolution of the instruments. The refined slopes, m_r, are given in Table 2. The slope obtained from the fit should reflect the instrument resolution. NPDF gives the shallowest slope; however, the correlation with resolution is not perfect. This is discussed more below.

As discussed in §3.2.1, the Q dependence of the instrument resolution is expected to result in an r-dependent broadening in the PDF peaks as well as a loss of intensity at high r. The

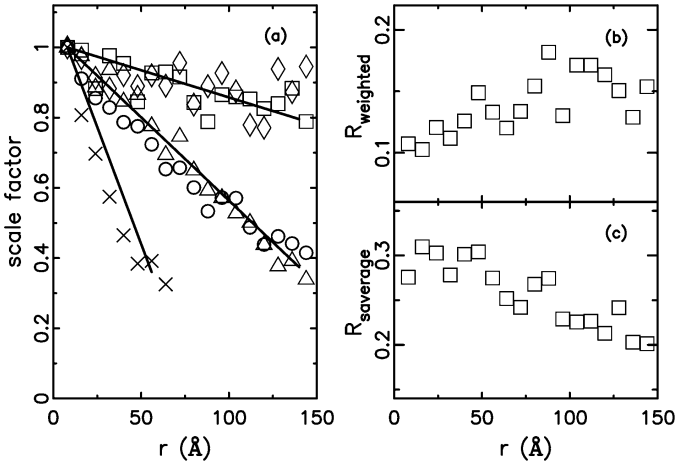


Figure 5
(a) Normalized scale factors, $S(r)/S(0)$, from 'box-car' PDF refinements from diffractometers NPDF (squares), NPD (diamonds), GEM (circles), GPPD (triangles) and GLAD (crosses). The linear fits to NPDF, GEM and GLAD are shown as solid lines. (b) Resulting R_{wp} from the 'box-car' refinements on NPDF data. (c) The mean square of the difference between the experimental and refined PDFs from the 'box-car' refinement.

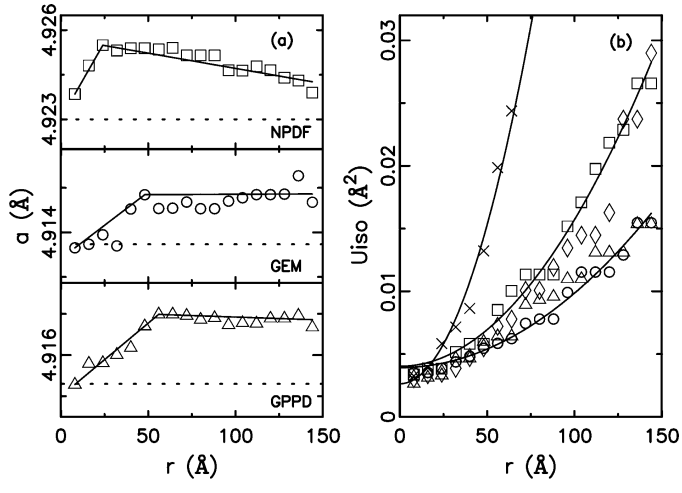


Figure 6
(a) Lattice constants from the 'box-car' refinements on NPDF, GEM and GPPD are shown in each panel as labelled. Rietveld-refined values are shown as dashed lines. Solid lines are guides to the eye. (b) Isotropic thermal factor from 'box-car' refinements with NPDF (squares), NPD (diamonds), GEM (circles), GPPD (triangles) and GLAD (crosses). Solid lines show the fits with Thorpe equation (1) on NPDF, GEM and GLAD data.

refined isotropic thermal factor *versus* r is shown in Fig. 6(*b*). Indeed the PDF peak widths do increase approximately quadratically as predicted. Both the Toby & Egami (1992) and Thorpe *et al.* (2002) forms fit the data well. In each case the Thorpe *et al.* (2002) form gives a better agreement, suggesting that the form of $\sigma_q^2(Q) = \sigma_{q0}^2 + \alpha^2 Q^2$ is quite an accurate description of the Q-dependent broadening of the IRF. In the model of Thorpe *et al.* (2002), σ_U^2 corresponds to the mean square amplitude of temperature vibrations. The fitted values of the temperature factor $U_{\rm PDF} = \sigma_U^2$ for Pb are listed in Table 2, together with their Rietveld-refined counterparts $U_{\rm Rw}$. It is clear in equation (1) that the refined value of α is related to the resolution, with the value of α smaller for the higher resolution machines. The data reflect this qualitatively, but not perfectly since α is smaller for GEM than NPDF, whereas their resolutions are similar or slightly better in the case of NPDF. Presumably, the reason is that in the PDFFIT refinements the damping factor (reflected in S) and the broadening factor (α) are somewhat correlated. The $G(r)$ function oscillates around zero and increasing the width of peaks has the effect of diminishing the amplitude of the oscillations, similar to a loss of intensity in the peaks. Comparing NPDF, NPD, GPPD and GEM, all of which have similar resolutions, we see that GPPD and GEM prefer a smaller $m_{\rm r}$ and a larger α to achieve essentially the same data damping as NPDF and NPD (as seen qualitatively in Fig. 3).

As discussed, the asymmetry of diffraction peaks results in aberrations of the lattice constants refined from the PDF. The r dependence of refined lattice parameters from NPDF, GEM and SEPD is shown in Fig. 6(*a*), where the dashed lines denote lattice parameters obtained by the Rietveld method. The PDF refinements consistently overestimate a, although these effects are rather small. In the case of NPDF, the average PDF-refined value overestimates a by $\sim$0.007 Å. This deviation is smaller than the differences in the Rietveld-refined lattice parameters from various instruments, which we ascribe to the approximately 10 K variability in the measurement temperatures. Nonetheless, it is clear that diffractometer correction parameters need to be included in refinement codes such as *PDFFIT* so that PDF- and Rietveld-derived lattice parameters from the same data agree.

The r dependence of the refined lattice parameters is somewhat small but significant, as evident in Fig. 6(*a*). It is also rather non-trivial. Refined lattice parameters increase approximately linearly at lower r, with an opposite downward, approximately linear trend at high r. There is also considerable variability between the diffractometers that does not depend simply on a single parameter such as resolution. For example, the SEPD results (not shown) are more highly peaked in the mid-r region than is the case for the higher resolution instruments. It may be necessary to refine diffractometer constants from a standard such as nickel over the refinement range of interest for a material of unknown lattice parameter. Alternatively, where appropriate, the lattice parameters could be fixed from a Rietveld refinement and diffractometer constants allowed to vary, reducing the residual without refining the physical parameter.

4. Conclusions

Analysis of the PDF round-robin experiment has verified that the finite measurement range directly affects the PDF real-space resolution, which increases with increasing $Q_{\rm max}$. The finite instrument resolution limits the probed PDF range in real space. The Q dependence of the instrument resolution results in the r dependence of the PDF real-space resolution, *i.e.* peak broadening at higher r. The PDF peak positions are found to shift slightly to higher r values than from the crystallographic analysis, and show some degree of r dependence. For a low-r PDF, the real-space resolution is not sensitive to the Q resolution of the instrument.

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References

Billinge, S. J. L. (1992). *Local Atomic Structure and Superconducivity of $Nd_{2-x}Ce_xCuO_{4-y}$: a Pair Distribution Function Study*, PhD thesis, University of Pennsylvania, USA.

Billinge, S. J. L., DiFrancesco, R. G., Kwei, G. H., Neumeier, J. J. & Thompson, J. D. (1996). *Phys. Rev. Lett.* **77**, 715–718.

Billinge, S. J. L. & Kanatzidis, M. G. (2003). *Chem. Commun.* Submitted.

Božin, E. S., Billinge, S. J. L., Takagi, H. & Kwei, G. H. (2000). *Phys. Rev. Lett.* **84**, 5856–5859.

Dutta, P. & Horn, P. M. (1981). *Rev. Mod. Phys.* **53**, 497–516.

Egami, T. & Billinge, S. J. L. (2003). *Underneath the Bragg Peaks: Structural Analysis of Complex Materials.* Oxford: Pergamon/Elsevier.

Hibble, S. J. & Hannon, A. C. (2002). *From Semiconductors to Proteins: Beyond the Average Structure*, edited by S. J. L. Billinge & M. F. Thorpe, *Fundamental Materials Research*, p. 129. New York: Kluwer Academic/Plenum.

Jeong, I.-K., Proffen, Th., Mohiuddin-Jacobs, F. & Billinge, S. J. L. (1999). *J. Phys. Chem. A*, **103**, 921–924.

Larson, A. C. & Von Dreele, R. B. (2000). *General Structure Analysis System*. Report No. LAUR-86-748, Los Alamos National Laboratory, Los Alamos, USA.

Louca, D., Egami, T., Brosha, E. L., Röder, H. & Bishop, A. R. (1997). *Phys. Rev. B*, **56**, R8475–R8478.

Mamontov, E. & Egami, T. (2000). *J. Phys. Chem. Solids*, **61**, 1345–1356.

Peterson, P. F., Gutmann, M., Proffen, Th. & Billinge, S. J. L. (2000). *J. Appl. Cryst.* **33**, 1192.

Peterson, P. F., Proffen, Th., Jeong, I.-K., Billinge, S. J. L., Choi, K.-S., Kanatzidis, M. G. & Radaelli, P. G. (2001). *Phys. Rev. B*, **63**, 165211.

Petkov, V., Billinge, S. J. L., Vogt, T., Ichimura, A. S. & Dye, J. L. (2002). *Phys. Rev. Lett.* **89**, 075502. (Highlighted in *Phys. Rev. Focus*: http://focus.aps.org/story/v10/st4.)

Proffen, Th. & Billinge, S. J. L. (1999). *J. Appl. Cryst.* **32**, 572–575.
Proffen, Th., Billinge, S. J. L., Egami, T. & Louca, D. (2003). *Z. Kristallogr.* **218**, 132–143.
Thorpe, M. F., Chung, J. S., Billinge, S. J. L. & Mohiuddin-Jacobs, F. (1998). *Local Structure from Diffraction*, edited by S. J. L. Billinge & M. F. Thorpe, p. 157. New York: Plenum.
Thorpe, M. F., Levashov, V. A., Lei, M. & Billinge, S. J. L. (2002). *From Semiconductors to Proteins: Beyond the Average Structure*, edited by S. J. L. Billinge & M. F. Thorpe, p. 105. New York: Kluwer/Plenum.
Toby, B. H. & Egami, T. (1992). *Acta Cryst.* A**48**, 336–346.
Tucker, M. G., Dove, M. T. & Keen, D. A. (2000). *J. Phys. Condens. Matter*, **12**, L723–L730.